PPI Tracker

Cell locking plan

Agreed behavior for single cells, full entity editing, and multiple selected rows | 10 November 2026

Goal: prevent two users from accidentally editing and overwriting the same tracker data at the same time.

Final rule

Edit in Cell = lock one cell.
Edit Entity dialog = lock one row.
Multi-row Edit Entity dialog = lock all selected rows together.

What starts a lock?

User action	System behavior
Click one cell or row	Select only. No lock is created.
Right-click -> Edit in Cell	Lock that single cell before opening the in-cell editor.
Double-click -> Edit Entity dialog	Lock the complete entity/row before opening the dialog.
Open Edit Entity with selected rows	Lock every selected row together before opening the bulk dialog.

Example: same row

User A double-clicks **anitaHero**. The server locks anitaHero and the Edit Entity dialog opens. If User B double-clicks anitaHero at the same time, the server rejects the second lock request and the dialog does not open for User B. The application shows: **“anitaHero is currently being edited by User A.”**

Example: same cell

User A right-clicks **Status** for anitaHero and selects Edit in Cell. Only Status is locked. User B cannot edit Status, but can edit a different cell such as Notes if no row lock exists.

How the feature will work

1. Single-cell editing

The application requests a cell lock after the user selects **Edit in Cell**. If the server grants it, the editor opens. On Save, Cancel, Escape, or editor close, the application releases the cell lock.

2. Full Edit Entity dialog

The dialog displays many editable fields, so it receives an entity/row lock. While User A has this dialog open, User B cannot open the full editor for the same row and cannot obtain a cell lock inside that row.

3. Multiple selected rows

The server locks selected rows as an all-or-nothing batch. If the selection contains anitaHero, bipedHero, and bookHero, all three must be available. If bipedHero is locked, the server locks none of them and the bulk dialog does not open.

4. Save, cancel, and crash behavior

Situation	Required behavior
Save	Check the lock and data revision. If valid, save data and release the lock.
Cancel or close	Release the lock without changing data.
Application crash or disconnect	The server automatically expires the lock after 2 to 5 minutes.
Editor remains open	The application renews the lock every 30 to 60 seconds.

5. Final safety check: revisions

Every save sends the revision number that was loaded when editing began. If another update changed the same field during editing, the revision no longer matches. The server rejects the save and shows a conflict instead of overwriting newer data.

Locks prevent normal simultaneous editing. Revision checks protect data if a lock expires, another client bypasses the lock, or an import updates the data.

Backend work to implement

Lock service: acquire, renew, and release cell and entity locks with expiry.
Batch locking: lock all selected rows or none of them.
Per-cell revisions: detect a stale edit during every save.
Atomic batch update: save all selected rows together or save none when a conflict exists.

Result

Users can select data freely. When they begin editing, PPI Tracker reserves the correct cell or row. Other users receive a clear message instead of opening a conflicting editor.